



Letting
event files and
BOLD runs
ghost each other



Solving
communication
issues
within datasets

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A Configurable QC Workflow for Task-fMRI Data Readiness

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Before analyzing brains, let's make sure
the files have "communicated."



Letting
event files and
BOLD runs
ghost each other

Solving
communication
issues
within datasets

Events
Behavior
Stimuli



Imaging
files



Config
file



QC
script



Report
+ WARNINGS

“Couples therapy” for task-fMRI files

A flexible workflow helps different datasets communicate clearly.

Linking event metadata with
imaging-file structure
before deeper analysis!

1. Introduce them!

→ Read event files and imaging-file listings

2. Check the relationship status...

→ Validate columns, file patterns, and event-BOLD matching

3. Document the drama :(

→ Write human-readable Markdown reports with warnings

Chen_project_2026 / bhs_task_fmri_qc /

x2583319 Add task-fMRI metadata QC open-data prototype

Name	Last commit message
..	
configs	
demo_data	
reports	
scripts	
README.md	Add task-fMRI metadata Q
requirements.txt	Add task-fMRI metadata Q

- dataset-specific expectations
- documented results
- reusable QC logic

Two datasets + Same QC

One configurable file-inventory check across two different task-fMRI datasets.

[Chen_project_2026](#) / [bhs_task_fmri_qc](#) / [reports](#) / [open_data_qc_comparison.md](#) 

 **x2583319** Add task-fMRI metadata QC open-data prototype

Preview Code Blame 17 lines (14 loc) · 570 Bytes

Open-data QC Comparison

Check	ds003720 music fMRI	ds000007 stop-signal task
Subjects	5	20
BOLD	90	118
Events	90	118
T1w	5	20
Events without BOLD	0	0
BOLD without Events	0	0

Interpretation

- ds003720 tests the workflow on a music fMRI dataset.
- ds000007 tests the workflow on a non-music task-fMRI dataset.
- The comparison focuses on file inventory and event/BOLD matching, not scientific analysis.
- Differences in event-table columns demonstrate why a configurable QC workflow is useful.

Takeaway

Both datasets had 0 events without BOLD and 0 BOLD runs without events.

The couples therapist leaves “receipts.”

A reproducible readiness check before deeper analysis.

1. The therapist

→ Snapshots of the matching code

```
Chen_project_2026 / bhs_task_fmri_qc / scripts / qc_openneuro_listing.py

Code Blame 153 lines (131 loc) · 5.53 KB

35 def main():
51
52     events = [k for k in keys if k.endswith("_events.tsv")]
53     bold = [k for k in keys if k.endswith("_bold.nii.gz") or k.endswith("_bold.nii")]
54     bold_json = [k for k in keys if k.endswith("_bold.json")]
55     t1w = [k for k in keys if k.endswith("_T1w.nii.gz") or k.endswith("_T1w.nii")]
56     fmap = [k for k in keys if "/fmap/" in k]

71 event_bases = {strip_suffix(k) for k in events}
72 bold_bases = {strip_suffix(k) for k in bold}
73
74 events_without_bold = sorted(event_bases - bold_bases)
75 bold_without_events = sorted(bold_bases - event_bases)
76
```

3. The status update

→ Relationship status:

- ☒ Communicating
- ☐ Ghosting
- ☐ “I thought that was the matching run”

2. The receipt

→ Smoke tests

OpenNeuro ds000007 Stop-signal Task Dataset Smoke-test QC Report

Scope

This is a remote file-inventory smoke test based on an AWS S3 listing. It checks file presence and BIDS-like matching between event files and BOLD files. It does not download or validate NIfTI image integrity.

Dataset-level summary

- Total files listed: 492
- Subjects detected: 20
- dataset_description.json found: PASS
- participants.tsv found: PASS

Image and event inventory

- BOLD NIfTI files: 118
- BOLD JSON sidecars: 122
- events.tsv files: 118
- T1w anatomical files: 20
- Fieldmap-related files: 0

Directory/modality counts

- func: 355
- anat: 80
- other/root: 57

Cross-check: events × BOLD

- Event bases detected: 118
- BOLD bases detected: 118
- Events without matching BOLD: 0
- BOLD without matching events: 0

Warnings

- PASS: No major file-inventory warnings detected in this scope.

Next step

Download a small subset, such as one subject/run, then add NIfTI readability checks with nibabel: shape, affine/header readability, voxel size, and number of BOLD volumes.

OpenNeuro ds003720 Music Genre fMRI Dataset Smoke-test QC Report

Scope

This is a remote file-inventory smoke test based on an AWS S3 listing. It checks file presence and BIDS-like matching between event files and BOLD files. It does not download or validate NIfTI image integrity.

Dataset-level summary

- Total files listed: 287
- Subjects detected: 5
- dataset_description.json found: PASS
- participants.tsv found: WARNING

Image and event inventory

- BOLD NIfTI files: 90
- BOLD JSON sidecars: 90
- events.tsv files: 90
- T1w anatomical files: 5
- Fieldmap-related files: 0

Directory/modality counts

- func: 270
- anat: 10
- other/root: 7

Cross-check: events × BOLD

- Event bases detected: 90
- BOLD bases detected: 90
- Events without matching BOLD: 0
- BOLD without matching events: 0

Warnings

- WARNING: participants.tsv not found.

Next step

Download a small subset, such as one subject/run, then add NIfTI readability checks with nibabel: shape, affine/header readability, voxel size, and number of BOLD volumes.

THANK YOU!

TLDR:

I made a reusable check for task-fMRI files.

It checks that task-event files and brain-scan runs are properly matched before anyone analyses the brains.

Because brains are complicated enough, but your filenames do not need to be.